

CloudShift - Software Management | SRS Document

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1. Introduction & Project Scope

1.1 Executive Summary & Purpose

This SRS document details functional, non-functional, interface, and operational requirements for **CloudShift**, migrating nine internal web applications from legacy on-premise hardware to a public cloud. The project modernizes system architecture while ensuring business continuity, data integrity, and zero-downtime cutover. By defining clear performance thresholds and validation metrics, this specification serves as an engineering contract and academic evaluation framework.

1.2 Scope of Infrastructure Modernization

The migration framework encompasses the following scope:

- **Application Artifact Transfer:** Migration of binaries, configurations, and relational database schemas for Applications 1–9.
- **Automated Infrastructure Deployment:** IaC pipelines, real-time data replication, traffic cutover protocols, and sub-15-minute rollbacks.
- **Business Continuity Enforcement:** Operational blackout windows during the final five working days of each month.
- **Legacy Risk Remediation:** Technical debt mitigation in undocumented legacy environments (App7, App8, App9).

1.3 Stakeholder Analysis & Elicitation Methodology

Requirements were gathered through a multi-method research and elicitation framework:

- **Legacy Document Analysis:** Auditing server configs, database mappings, and static code across App1–App6.
- **Structured Stakeholder Workshops:** Design sessions with leadership, Lead Infrastructure Engineers, DevOps, and Business Units.
- 1. **Historical Post-Mortem Analysis:** Evaluating prior billing outage data to establish non-functional guardrails.

2. System Constraints & Operational Environment

2.1 Critical System Constraints

- **C-01 (Hardware End-of-Life Boundary):** On-premise hardware support expires in seven months. Cutover must finish prior to termination.
- **C-02 (Operational Blackout Protocol):** Automated deployment pipelines and cutover procedures are locked during the last five working days monthly.
- **C-03 (Human Resource Ceiling):** Engineering work is capped at a core team of three personnel.
- **C-04 (Legacy Codebase Vulnerability):** App7, App8, and App9 lack documentation, requiring dynamic discovery techniques.

3. Core Findings & Functional Requirements (FR)

The system must satisfy twelve core Functional Requirements (FR-01 through FR-12) for operational readiness:

Requirement ID	Requirement Name	Description	Stakeholder Origin
FR-01	Data Sync & Staging	The system shall continuously replicate database transactions from on-premise databases to staging cloud databases in real time using Change Data Capture (CDC).	Cloud Engineer / DevOps
FR-02	Schema Migration	The system shall automatically transform legacy database schemas into target cloud-native database engine formats without structural data loss.	Database Administrator
FR-03	Traffic Rerouting & DNS Cutover	The system shall update DNS routing tables and load balancer target groups to direct end-user application traffic from on-premise endpoints to cloud endpoints.	Network Admin / CIO
FR-04	Blackout Enforcement	The system shall automatically lock all automated deployment pipelines and prevent cutover execution during the designated last 5 working days of each month.	Business Teams / Billing
FR-05	Session Persistence	The system shall migrate active user sessions to a centralized cloud session cache (e.g., Redis cluster) prior to	Business User / UX Lead

		cutover to prevent session drop-outs.	
FR-06	Automated IaC Provisioning	The system shall deploy cloud compute, networking, storage, and security groups using declarative Infrastructure-as-Code (IaC) scripts.	Cloud Engineer
FR-07	Config Management	The system shall externalize application configuration parameters into a centralized cloud parameter store, injecting environment secrets dynamically at launch.	Security Lead / Engineer
FR-08	Cloud IAM & RBAC	The system shall enforce role-based access control (RBAC) integrated with corporate SSO (Single Sign-On) for all 9 application administrative interfaces.	Security Administrator
FR-09	Synthetic Health Monitoring	The system shall perform automated synthetic transaction checks against staging and production endpoints every 30 seconds to verify functional health.	System Monitor / Operations
FR-10	Automated Rollback Trigger	The system shall automatically revert DNS traffic back to on-premise infrastructure within 15 minutes if post-cutover synthetic checks detect failure threshold breaches.	CIO / Business Teams
FR-11	Audit Logging & Compliance	The system shall log all migration actions, cutover triggers, configuration edits, and user access events to an immutable append-only cloud audit storage.	Compliance / Auditor
FR-12	Real-Time Migration Dashboard	The system shall provide a unified management dashboard	Project Manager / CIO

		showing real-time replication lag, cutover status, blackout window countdown, and application health.	
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4. Non-Functional Requirements (NFR) & Measurable Metrics

Every non-functional requirement contains explicit, empirical metrics rather than qualitative statements:

NFR ID	Category	Requirement Description	Measurable Target / Metric	Verification Method
NFR-01	Availability	Overall Cloud Infrastructure Uptime	Post-migration cloud availability shall be $\geq 99.95\%$ per calendar month (max unplanned downtime ≤ 21.9 minutes/month).	CloudWatch / SLA Monitor
NFR-02	Availability	Maximum Cutover Downtime Window	Unplanned downtime during application cutover shall not exceed 30 minutes per application during scheduled maintenance.	Downtime Timer / Log Audit
NFR-03	Performance	API Response Latency	95th percentile (P95) HTTP API response time shall be $\leq 200\text{ ms}$ under a load of 1,000 concurrent requests/sec.	JMeter Load Test
NFR-04	Performance	Database Query Latency	99th percentile (P99) transaction execution time for billing/dispatch queries shall be $\leq 50\text{ ms}$.	APM Query Profiler
NFR-05	Data Integrity	Recovery Point Objective (RPO)	Data loss during migration cutover shall be RPO = 0 seconds (zero transaction loss).	CDC Audit Log Comparison
NFR-06	Data Integrity	Pre-Cutover Checksum Validation	100% data integrity parity match (SHA-256)	Hash Verification Script

			hash verification) between on-premise source DB and target DB before cutover authorization.	
NFR-07	Security	Data Encryption Standards	100% of data at rest encrypted with AES-256 and all data in transit encrypted using TLS 1.3.	Cloud Security Scanner
NFR-08	Maintainability	IaC Deployment Validation	100% of infrastructure changes deployed via IaC pipelines with continuous automated linting and zero manual configuration drift.	IaC Pipeline Linter & Drift Audit

5. Architectural Prioritization (MoSCoW Framework)

To balance technical complexity with resource limits, requirements follow MoSCoW prioritization [Verify Citation]:

5.1 MUST HAVE (Critical Baseline Requirements)

- **Functional Mandatory Core:** FR-01 (Data Synchronization), FR-03 (DNS Cutover), FR-04 (Blackout Locking), FR-06 (IaC Provisioning), and FR-10 (Automated Rollback Trigger).
- **Non-Functional Mandatory Core:** NFR-02 (Cutover Downtime $\leq 30m$), NFR-05 (RPO = 0s), NFR-06 (SHA-256 Parity Match), and NFR-08 (RTO $\leq 15m$).

5.2 SHOULD HAVE (High-Priority Architectural Enhancements)

- **Functional Enhancements:** FR-02 (Schema Transformation), FR-05 (Session Persistence), FR-07 (Secrets Management), FR-08 (SSO/RBAC Security), FR-09 (Synthetic Monitoring), and FR-11 (Immutable Audit Logging).
- **Non-Functional Enhancements:** NFR-01 (99.95% Availability), NFR-03 (P95 Latency $\leq 200ms$), and NFR-04 (P99 Query Latency $\leq 50ms$).

5.3 COULD HAVE (Desirable Non-Critical Capabilities)

- FR-12 (Real-Time Executive Migration Dashboard) and automated continuous scanning for TLS 1.3 / AES-256 validation (NFR-07).

5.4 WON'T HAVE (Explicit Out-of-Scope Items for Initial Release)

Full microservices refactoring, serverless rewrites, and non-essential feature development during the initial 7-month lifecycle.

6. Requirements Traceability Matrix (RTM)

The Traceability Matrix maps functional requirements to quality targets, modules, and test cases:

FR ID	Functional Requirement Summary	Related NFR	Architecture Module	Test Case ID
FR-01	Data Sync & Staging	NFR-05, NFR-06	Data Pipeline / CDC Engine	TC-FR-01
FR-02	Schema Migration	NFR-05, NFR-06	Database Transformer	TC-FR-02
FR-03	Traffic Rerouting & DNS Cutover	NFR-02	Traffic Orchestrator / DNS	TC-FR-03
FR-04	Blackout Enforcement	NFR-02	Blackout Guard Service	TC-FR-04
FR-05	Session Persistence	NFR-03	Session State Cache	TC-FR-05
FR-06	Automated IaC Provisioning	NFR-01	Terraform IaC Modules	TC-FR-06
FR-07	Config Management	NFR-07	Cloud Secrets Manager	TC-FR-07
FR-08	Cloud IAM & RBAC	NFR-07	Security IAM Service	TC-FR-08
FR-09	Synthetic Health Monitoring	NFR-01, NFR-03	Synthetic Monitor Bot	TC-FR-09
FR-10	Automated Rollback Trigger	NFR-08	Rollback Controller	TC-FR-10
FR-11	Audit Logging & Compliance	NFR-07	Audit Logger Service	TC-FR-11
FR-12	Real-Time Migration Dashboard	NFR-01	Management UI Console	TC-FR-12

7. Discussion & Implementation Analysis

The technical specifications strictly align with the case study parameters [Verify Citation]. Quantitative targets in Section 4 serve as binding acceptance criteria. Each metric includes a corresponding verification protocol for post-deployment validation. Architecturally, the strategy emphasizes deterministic failure modes, automated health verification, and low-latency failovers over manual intervention.

Capacity Analysis & Constraint Interpretation: The seven-month contract expiration creates a firm deadline. Enforcing the monthly 5-day blackout across 22 working days leaves an effective window of 17 working days monthly. Execution throughput must be optimized via automated IaC pipelines and continuous data replication to eliminate manual overhead.

MoSCoW prioritization ensures security standards—AES-256 encryption at rest and TLS 1.3 in transit—are mandatory operational guardrails. Secondary tools like the executive dashboard remain lower priority, focusing engineering effort on replication integrity, failure detection, and deterministic cutovers.

8. Conclusion & Next Steps

Project CloudShift provides a structured path to eliminate hardware end-of-life risks while preserving operational continuity. Through CDC synchronization, automated rollbacks, and strict blackout enforcement, the system minimizes business disruption. Next steps involve finalizing staging scripts, validating synthetic check suites, and executing initial mock cutovers before final production execution [Insert Date Here].